AERO50008 – Structures 2: Buckling of struts

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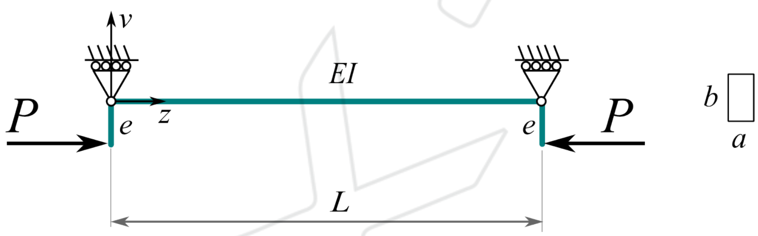
(**1.**) Assume the imperfect strut with eccentricity e, in the line of action of the load.

i. Find its critical buckling load.

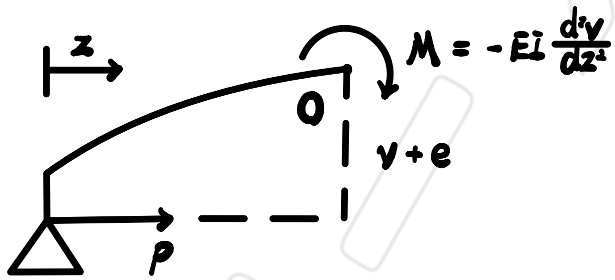
ii. How does the eccentricity influence the buckling load?

iii. What is the influence of the eccentricity upon the maximum deflection? Plot the ratio load P to critical Euler buckling ratio *vs.* maximum deflection.

iv. Calculate the buckling load for a beam 10 m long, made of steel ($E = 210000$ MPa, $\nu = 0.3$), rectangular cross-section ($a = 100$ mm, $b = 400$ mm), with 100 mm eccentricity in the loading.



i)



When reach the buckling load. at O

$$M = -EI\frac{d^2v}{dz^2} = P(v+e) = Pv + Pe$$

$$\therefore EI\frac{d^2v}{dz^2} + Pv = -Pe \qquad \therefore \frac{d^2v}{dz^2} + \mu^2 v = -\mu^2 e \quad \text{where } \mu^2 = \frac{P}{EI}$$

$\therefore v = A\sin(\mu z) + B\cos(\mu z) + C$

$\therefore \frac{d^2v}{dz^2} = -\mu^2[A\sin(\mu z) + B\cos(\mu z)]$

$\therefore \frac{d^2v}{dz^2} + \mu^2 v = \mu^2 C = -\mu^2 e \qquad \therefore C = -e$

$\therefore v = A\sin(\mu z) + B\cos(\mu z) - e$

At $z = 0$, $v = 0$

$B - e = 0 \qquad \therefore B = e \qquad \therefore v = A\sin(\mu z) + e[\cos(\mu z) - 1]$

At $z = L$, $v = 0$

$\therefore A\sin(\mu L) + e[\cos(\mu L) - 1] = 0$

$\because A \neq 0 \qquad \therefore \sin(\mu L) = \cos(\mu L) - 1 = 0 \qquad \therefore \mu L = 0$ or $n\pi$ $(n \in \mathbb{R})$

$\therefore \mu L = \pi \qquad \therefore \mu^2 L^2 = \frac{P}{EI} L^2 = \pi^2 \qquad \therefore P_{crit} = \frac{\pi^2 EI}{L^2}$

ii) P_{crit} is not change

iii) $A = \frac{-e[\cos(\mu L) - 1]}{\sin(\mu L)}$

At $z = \frac{L}{2}$,

$$v_c = \frac{-e[\cos(\mu L) - 1]}{\sin(\mu L)} \sin(\tfrac{1}{2}\mu L) + e\cos(\tfrac{1}{2}\mu L) - e$$

$$= \frac{2e\sin(\frac{1}{2}\mu L)\sin(\frac{1}{2}\mu L)}{2\sin(\frac{1}{2}\mu L)\cos(\frac{1}{2}\mu L)} \sin(\tfrac{1}{2}\mu L) + e\cos(\tfrac{1}{2}\mu L) - e$$

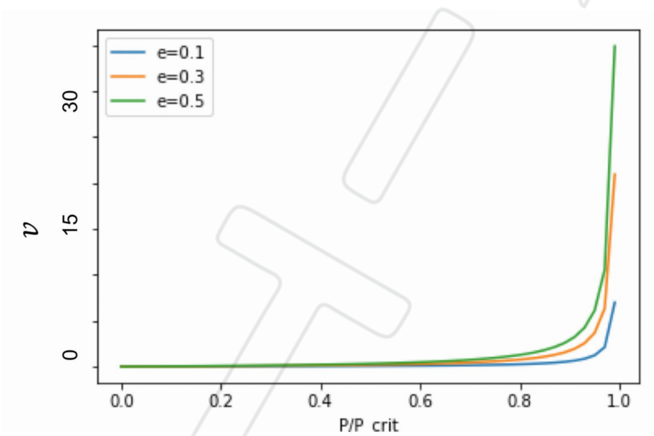
$$= e\frac{\sin^2(\frac{1}{2}\mu L)}{\cos(\frac{1}{2}\mu L)} + e\frac{\cos^2(\frac{1}{2}\mu L)}{\cos(\frac{1}{2}\mu L)} - e \quad = e\left[\frac{1}{\cos(\frac{1}{2}\mu L)} - 1\right]$$

$$\because \mu = \sqrt{\frac{P}{EI}} = \sqrt{\frac{P_{crit}}{EI}\frac{P}{P_{crit}}} = \sqrt{\frac{\pi^2 EI}{L^2}\cdot\frac{1}{EI}}\cdot\sqrt{\frac{P}{P_{crit}}} = \frac{\pi}{L}\sqrt{\frac{P}{P_{crit}}}$$

$$\therefore v_c = e\left[\frac{1}{\cos(\frac{\pi}{2}\sqrt{\frac{P}{P_{crit}}})} - 1\right]$$

$\therefore v_c$ tend to be ∞ as $P_{crit} \longrightarrow P$

$$\therefore \frac{P}{P_{crit}} = \left[\frac{2}{\pi}\cos^{-1}\left(\frac{e}{v_c + e}\right)\right]^2 = \left[\frac{2}{\pi}\cos^{-1}\left(\frac{e}{v_c}\right)\right]^2 \text{ for small } e$$



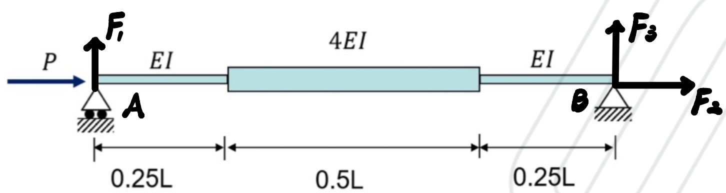
* ∞ at $P = P_{crit}$

iv) $P_{crit} = \dfrac{\pi^2 EI}{L^2} = 1.1054 \times 10^8\ N$

(**2.**) Consider the pin-ended strut shown below.

i. Obtain the transcendental equation (i.e. displacement at any point) for the whole strut.

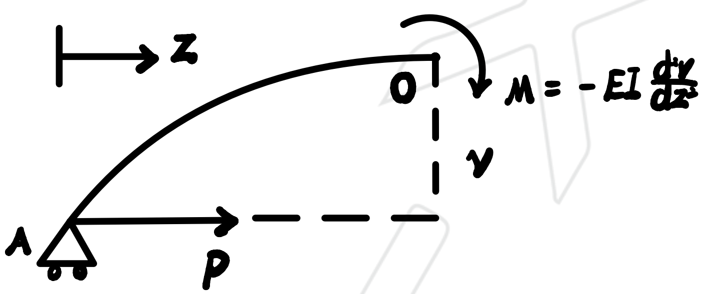
ii. Find the critical buckling load for the structure.



$$\Sigma F_v = F_1 + F_3 = 0$$

$$\Sigma F_h = F_2 + P = 0 \qquad \therefore F_2 = -P$$

At A, $M(A) = F_3 L = 0 \qquad \therefore F_3 = 0 \qquad \therefore F_1 = 0$



At O, $\Sigma M = -M + Pv = EI\frac{d^2v}{dz^2} + Pv = 0$

for $0 \leq z \leq \frac{1}{4}L$ and $\frac{3}{4}L \leq z \leq L$,

$\therefore \frac{d^2v}{dz^2} + \mu^2 v = 0$ where $\mu^2 = \frac{P}{EI}$

$\therefore v_1 = A\sin(\mu z) + B\cos(\mu z)$

for $\frac{1}{4}L \le z \le \frac{3}{4}L$.

$\therefore \frac{d^2v}{dz^2} + \frac{1}{4}\mu^2 v = 0$ where $\mu^2 = \frac{P}{EI}$

$\therefore v_2 = C\sin(\frac{1}{2}\mu z) + D\cos(\frac{1}{2}\mu z)$

At $z = 0$, $v_1 = 0$

$\therefore B = 0$ $\quad\therefore v_1 = A\sin(\mu z)$

At $z = \frac{1}{2}L$, $\frac{dv_2}{dz} = 0$

$\therefore \frac{1}{2}\mu[C\cos(\frac{1}{4}\mu L) - D\sin(\frac{1}{4}\mu L)] = 0 \quad \because \mu \neq 0$

$\therefore C = D\tan(\frac{1}{4}\mu L)$

$\therefore v_2 = D\sin(\frac{1}{2}\mu z)\tan(\frac{1}{4}\mu L) + D\cos(\frac{1}{2}\mu z)$

$= D[\sin(\frac{1}{2}\mu z)\tan(\frac{1}{4}\mu L) + \cos(\frac{1}{2}\mu z)]$

At $z = \frac{1}{4}L$, $v_1 = v_2$, $\frac{dv_1}{dz} = \frac{dv_2}{dz}$

$$\therefore A\sin\left(\tfrac{1}{4}\mu L\right) = D\left[\sin\left(\tfrac{1}{8}\mu L\right)\tan\left(\tfrac{1}{4}\mu L\right) + \cos\left(\tfrac{1}{8}\mu L\right)\right]$$

$$= D\left[\sin\left(\tfrac{1}{8}\mu L\right)\frac{\sin\left(\frac{1}{4}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)} + \cos\left(\tfrac{1}{8}\mu L\right)\right]$$

$$= D\left[\frac{2\sin^2\left(\frac{1}{8}\mu L\right)\cos\left(\frac{1}{8}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)} + \cos\left(\tfrac{1}{8}\mu L\right)\right]$$

$$= D\cos\left(\tfrac{1}{8}\mu L\right)\left[\frac{2\sin^2\left(\frac{1}{8}\mu L\right) + \cos\left(\frac{1}{4}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)}\right]$$

$$= D\cos\left(\tfrac{1}{8}\mu L\right)\left[\frac{1 - \cos\left(\frac{1}{4}\mu L\right) + \cos\left(\frac{1}{4}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)}\right]$$

$$= D\cos\left(\tfrac{1}{8}\mu L\right)\sec\left(\tfrac{1}{4}\mu L\right)$$

$$\mu A\cos\left(\tfrac{1}{4}\mu L\right) = \tfrac{1}{2}\mu D\left[\cos\left(\tfrac{1}{8}\mu L\right)\tan\left(\tfrac{1}{4}\mu L\right) - \sin\left(\tfrac{1}{8}\mu L\right)\right]$$

$$= \tfrac{1}{2}\mu D\left[\cos\left(\tfrac{1}{8}\mu L\right)\frac{2\sin\left(\frac{1}{8}\mu L\right)\cos\left(\frac{1}{8}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)} - \sin\left(\tfrac{1}{8}\mu L\right)\right]$$

$$= \tfrac{1}{2}\mu D\sin\left(\tfrac{1}{8}\mu L\right)\left[\frac{2\cos^2\left(\frac{1}{8}\mu L\right) - \cos\left(\frac{1}{4}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)}\right]$$

$$= \tfrac{1}{2}\mu D\sin\left(\tfrac{1}{8}\mu L\right)\left[\frac{1 + \cos\left(\frac{1}{4}\mu L\right) - \cos\left(\frac{1}{4}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)}\right]$$

$$= \tfrac{1}{2}\mu D\sin\left(\tfrac{1}{8}\mu L\right)\sec\left(\tfrac{1}{4}\mu L\right)$$

$$\therefore A = D\cos\left(\tfrac{1}{8}\mu L\right)\sec\left(\tfrac{1}{4}\mu L\right)\operatorname{cosec}\left(\tfrac{1}{4}\mu L\right)$$

$$= \tfrac{1}{2} D \sin\left(\tfrac{1}{8}\mu L\right)\sec^2\left(\tfrac{1}{4}\mu L\right)$$

$$\therefore \cos\left(\tfrac{1}{8}\mu L\right)\operatorname{cosec}\left(\tfrac{1}{4}\mu L\right) = \tfrac{1}{2}\sin\left(\tfrac{1}{8}\mu L\right)\sec\left(\tfrac{1}{4}\mu L\right)$$

$$\therefore \tfrac{1}{2} = \frac{1}{\tan\left(\tfrac{1}{8}\mu L\right)\cdot\tan\left(\tfrac{1}{4}\mu L\right)}$$

$$\therefore \tfrac{1}{2}\tan\left(\tfrac{1}{4}\mu L\right) = \frac{1}{\tan\left(\tfrac{1}{8}\mu L\right)}$$

$$\because \tan(2x) = \frac{2\tan(x)}{1-\tan^2(x)}$$

$$\therefore \tan\left(\tfrac{1}{4}\mu L\right) = \frac{2\tan\left(\tfrac{1}{8}\mu L\right)}{1-\tan^2\left(\tfrac{1}{8}\mu L\right)}$$

$$\therefore \frac{\tan\left(\tfrac{1}{8}\mu L\right)}{1-\tan^2\left(\tfrac{1}{8}\mu L\right)} = \frac{1}{\tan\left(\tfrac{1}{8}\mu L\right)}$$

$$\therefore \tan^2\left(\tfrac{1}{8}\mu L\right) = \tfrac{1}{2} \qquad \therefore \tan\left(\tfrac{1}{8}\mu L\right) = \tfrac{1}{\sqrt{2}} \qquad \therefore \tfrac{1}{8}\mu L = 0.6155$$

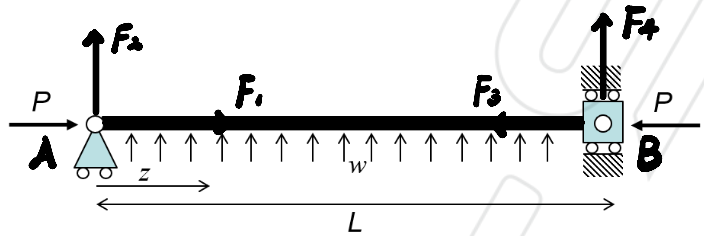
$$\therefore \mu L = 4.9238 \qquad \therefore \mu^2 L^2 = \frac{P}{EI}L^2 = 24.2442$$

$$\therefore P_{crit} = \frac{24.2442\,EI}{L^2}$$

(**3.**) Consider the pin-ended strut shown in the figure, subjected to an axial load P and a uniformly distributed lateral force of intensity w. Derive the expressions for the mid-span deflection and the critical buckling load, and discuss how this is influenced by the lateral load.

i. Formulate the differential equation to assess stability of the system.

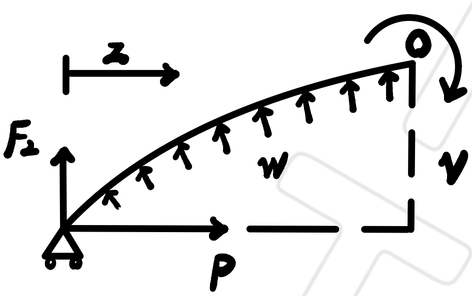
ii. Find the critical buckling load for the structure.



$$\Sigma F_v = F_2 + F_4 = -wL$$

$$\Sigma F_h = F_1 + F_3 = 0 \qquad \therefore F_1 = -F_3 = 0$$

$$M(A): \quad F_4 L + \frac{1}{2}wL^2 = 0 \qquad \therefore F_4 = -\frac{1}{2}wL \qquad \therefore F_2 = -\frac{1}{2}wL$$



$$M = -EI\frac{d^2v}{dz^2}$$

$$\text{At } O: \quad \Sigma M = EI\frac{d^2v}{dz^2} - \frac{1}{2}wz^2 + Pv + \frac{1}{2}wLz = 0$$

$$\therefore EI\frac{d^2v}{dz^2} + Pv = \frac{1}{2}wz^2 - \frac{1}{2}wLz$$

$$\therefore \frac{d^2v}{dz^2} + \mu^2 v = \frac{wz^2}{2EI} - \frac{wLz}{2EI} \quad \text{where} \quad \mu^2 = \frac{P}{EI}$$

$$\therefore v = A\sin(\mu z) + B\cos(\mu z) + Cz^2 + Dz + E$$

$$\therefore \frac{d^2v}{dz^2} = -\mu^2[A\sin(\mu z) + B\cos(\mu z)] + 2C$$

$$\therefore \mu^2 Cz^2 + \mu^2 Dz + \mu^2 E + 2C = \frac{wz^2}{2EI} - \frac{wLz}{2EI}$$

$$\therefore C = \frac{w}{2EI\mu^2} = \frac{w}{2P}$$

$$D = \frac{-wL}{2EI\mu^2} = -\frac{wL}{2P}$$

$$E = -\frac{2C}{\mu^2} = -\frac{w}{P\mu^2}$$

$$\therefore v = A\sin(\mu z) + B\cos(\mu z) + \frac{w}{2P}z^2 - \frac{wL}{2P}z - \frac{w}{P\mu^2}$$

At $z = 0,\ v = 0$

$$\therefore B - \frac{w}{P\mu^2} = 0 \qquad \therefore B = \frac{w}{P\mu^2}$$

At $z = L$, $v = 0$

$$\therefore A\sin(\mu L) + \frac{W}{P\mu^2}\cos(\mu L) - \frac{W}{P\mu^2} = 0$$

$$\therefore A\sin(\mu L) + \frac{W}{P\mu^2}[\cos(\mu L) - 1] = 0$$

$$\because A \neq 0 \qquad \therefore A = -\frac{W}{P\mu^2}\frac{\cos(\mu L) - 1}{\sin(\mu L)}$$

$$\therefore \sin(\mu L) = \cos(\mu L) - 1 = 0$$

$$\therefore \mu L = 0 \text{ or } n\pi \ (n \in \mathbb{R}) \qquad \therefore \mu L = \pi$$

$$\therefore \mu^2 L^2 = \pi^2 = \frac{P}{EI}L^2$$

$$\therefore P_{crit} = \frac{\pi^2 EI}{L^2}$$

At $z = \frac{L}{2}$, $v_c = v_{max}$

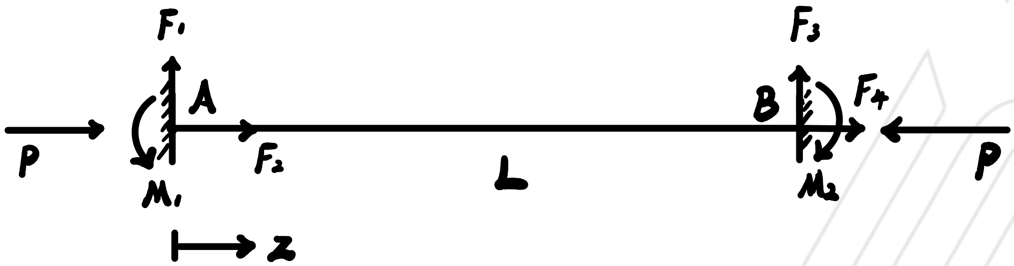
$$\therefore v_c = -\frac{W}{P\mu^2}\frac{\cos(\mu L) - 1}{\sin(\mu L)}\sin(\tfrac{1}{2}\mu L) + \frac{W}{P\mu^2}\cos(\tfrac{1}{2}\mu L) + \frac{W}{8P}L^2 - \frac{WL}{4P}L - \frac{W}{P\mu^2}$$

$$= \frac{W}{P\mu^2}\left[\frac{2\sin^2(\frac{1}{2}\mu L)}{2\sin(\frac{1}{2}\mu L)\cos(\frac{1}{2}\mu L)}\sin(\tfrac{1}{2}\mu L) + \cos(\tfrac{1}{2}\mu L) - 1\right] - \frac{W}{8P}L^2$$

$$\therefore v_c = \frac{W}{P\mu^2}\left[\frac{\sin^2(\tfrac{1}{2}\mu L) + \cos^2(\tfrac{1}{2}\mu L)}{\cos(\tfrac{1}{2}\mu L)} - 1\right] - \frac{W}{8P}L^2$$

$$= \frac{W}{P\mu^2}\left[\sec(\tfrac{1}{2}\mu L) - 1\right] - \frac{W}{8P}L^2$$

(**4.**) Calculate the buckling load of a fixed-fixed strut.



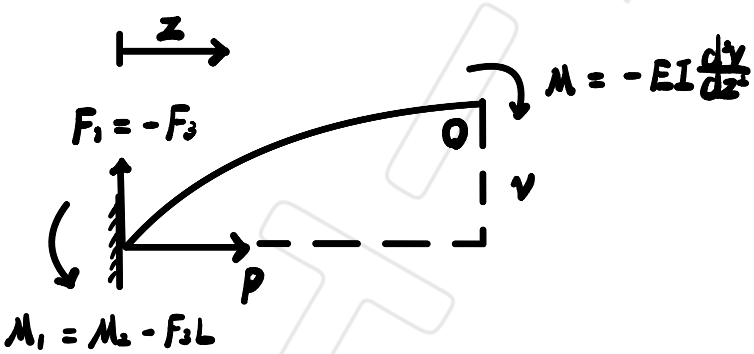
$$\Sigma F_v = F_1 + F_3 = 0 \qquad \therefore F_1 = -F_3$$

$$\Sigma F_h = F_2 + F_4 = 0 \qquad \therefore F_2 = F_4 = 0$$

$$\text{At A, } \Sigma M = M_1 + F_3 L - M_2 = 0$$

$$\text{At B, } \Sigma M = M_2 + F_1 L - M_1 = 0$$

$$\therefore M_1 = M_2 - F_3 L$$



$$\text{At O, } \Sigma M = EI\frac{d^2v}{dz^2} + Pv + F_3 z + M_2 - F_3 L$$

$$\therefore \frac{d^2v}{dz^2} + \mu^2 v = \frac{F_3}{EI}(L-z) - \frac{M_2}{EI} \quad \text{where} \quad \mu^2 = \frac{P}{EI}$$

$$\therefore v = A\sin(\mu z) + B\cos(\mu z) + Cz + D$$

$$\therefore \frac{d^2v}{dz^2} = -\mu^2[A\sin(\mu z) + B\cos(\mu z)]$$

$$\therefore \frac{d^2v}{dz^2} + \mu^2 v = \mu^2 Cz + \mu^2 D = \frac{F_3}{EI}(L-z) - \frac{M_2}{EI} = -\frac{F_3}{EI}z + \frac{F_3}{EI}L - \frac{M_2}{EI}$$

$$\therefore C = -\frac{F_3}{EI\mu^2} = -\frac{F_3}{P}$$

$$D = \frac{F_3}{EI\mu^2}L - \frac{M_2}{EI\mu^2} = \frac{F_3}{P}L - \frac{M_2}{P}$$

$$\therefore v = A\sin(\mu z) + B\cos(\mu z) - \frac{F_3}{P}z + \frac{F_3}{P}L - \frac{M_2}{P}$$

At $z=0$, $v=0$ and $\frac{dv}{dz}=0$

$$\therefore B + \frac{F_3}{P}L - \frac{M_2}{P} = 0 \quad \text{and} \quad \mu A - \frac{F_3}{P} = 0$$

$$\therefore B = -\frac{F_3}{P}L + \frac{M_2}{P} \quad \text{and} \quad A = \frac{F_3}{P\mu}$$

$$\therefore v = \frac{F_3}{P\mu}\sin(\mu z) + \left(\frac{M_2}{P} - \frac{F_3}{P}L\right)\cos(\mu z) - \frac{F_3}{P}z + \frac{F_3}{P}L - \frac{M_2}{P}$$

At $z=L$, $v=0$ and $\frac{dv}{dz}=0$

$$\therefore \frac{F_3}{P\mu}\sin(\mu L) + \left(\frac{M_2}{P} - \frac{F_3}{P}L\right)\left[\cos(\mu L) - 1\right] - \frac{F_3}{P}L = 0$$

and $$\frac{F_3}{P}\cos(\mu L) - \mu\left(\frac{M_2}{P} - \frac{F_3}{P}L\right)\sin(\mu L) - \frac{F_3}{P} = 0$$

$$\because F_1 = -F_3 = 0$$

$$\therefore \frac{M_2}{P}\left[\cos(\mu L) - 1\right] = 0 \quad \text{and} \quad \mu\frac{M_2}{P}\sin(\mu L) = 0$$

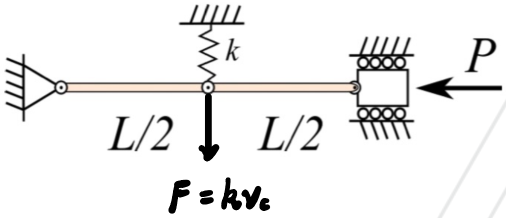
$$\therefore \sin(\mu L) = 0 \quad \text{while} \quad \cos(\mu L) = 1$$

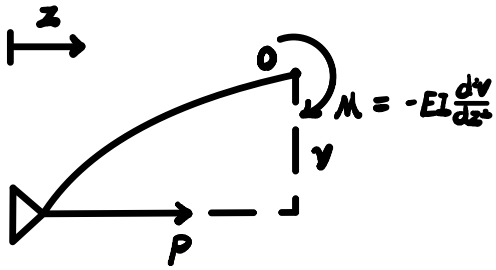
$$\therefore \mu L = 0 \text{ or } 2n\pi,\ n \in \mathbb{R} \qquad \therefore \mu L = 2\pi$$

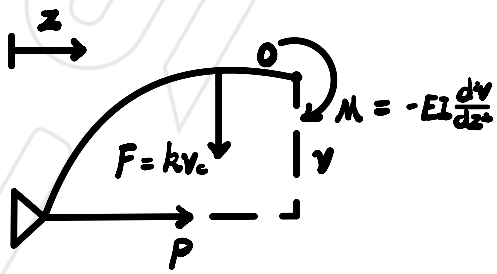
$$\therefore \mu^2 L^2 = 4\pi^2 = \frac{P}{EI}L^2$$

$$\therefore P = \frac{4\pi^2 EI}{L^2}$$

(**5.**) The structure shown below consists of two rigid members of length $L/2$ connected via pins. Use k as the spring's stiffness constant. What is the critical buckling load?







for $0 \le z \le \frac{1}{2}L$, $\mu^2 = \frac{P}{EI}$

$$\Sigma M(0) = EI\frac{d^2v}{dz^2} + Pv = 0 \qquad \therefore \frac{d^2v}{dz^2} + \mu^2 v = 0$$

$$\therefore v_1 = A\sin(\mu z) + B\cos(\mu z)$$

for $\frac{1}{2}L \le z \le L$, $\mu^2 = \frac{P}{EI}$

$$\Sigma M(0) = EI\frac{d^2v}{dz^2} + Pv + kV_c(z - \tfrac{1}{2}L) = 0 \qquad \therefore \frac{d^2v}{dz^2} + \mu^2 v = \frac{kV_c}{EI}(\tfrac{1}{2}L - z)$$

$$\therefore v_2 = C\sin(\mu z) + D\cos(\mu z) + Ez + F$$

$$\therefore \frac{d^2v_2}{dz^2} = -\mu^2[C\sin(\mu z) + D\cos(\mu z)]$$

$$\therefore Ez + F = \frac{kV_c}{EI}\left(\frac{1}{2}L - z\right)$$

$$\therefore v_2 = C\sin(\mu z) + D\cos(\mu z) - \frac{kV_c}{EI}z + \frac{kV_c}{2EI}L$$

At $z = 0$, $v = 0$

$$\therefore v_1 = B = 0 \qquad \therefore v_1 = A\sin(\mu z)$$

At $z = \frac{1}{2}L$, $v_1 = v_2$, $\frac{dv_1}{dz} = \frac{dv_2}{dz}$

$$\therefore A\sin\left(\tfrac{1}{2}\mu L\right) = C\sin\left(\tfrac{1}{2}\mu L\right) + D\cos\left(\tfrac{1}{2}\mu L\right)$$

$$\mu A\cos\left(\tfrac{1}{2}\mu L\right) = \mu C\cos\left(\tfrac{1}{2}\mu L\right) - \mu D\sin\left(\tfrac{1}{2}\mu L\right) - \frac{kV_c}{EI}$$

$$\therefore C + D\cot\left(\tfrac{1}{2}\mu L\right) = C - D\tan\left(\tfrac{1}{2}\mu L\right) - \frac{kV_c}{EI\mu}\sec\left(\tfrac{1}{2}\mu L\right)$$

$$\therefore D\left[\cot\left(\tfrac{1}{2}\mu L\right) + \tan\left(\tfrac{1}{2}\mu L\right)\right] + \frac{kV_c}{EI\mu}\sec\left(\tfrac{1}{2}\mu L\right) = 0$$

At $z = L$, $v_2 = 0$

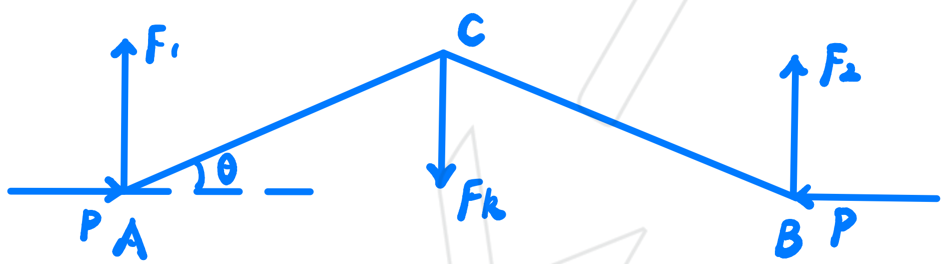
$$\therefore C\sin(\mu L) + D\cos(\mu L) - \frac{kV_c}{2EI}L = 0$$

$$\Sigma F_v = F_1 + F_2 = F_k$$

$$\Sigma M_A = F_2 L - F_k\left(\tfrac{1}{2}L\right) = 0 \qquad \therefore\ F_2 = \tfrac{1}{2}F_k = F_1$$

$$F_k = k \cdot L = k \cdot y_c = k \cdot \tfrac{1}{2}L\sin\theta = \tfrac{1}{2}kL\theta$$

$$\Downarrow$$



At A, $\Sigma F_v = F_1 + T_{AC}\sin\theta = 0$

$$\Sigma F_h = P + T_{AC}\cos\theta = 0$$

$$\therefore\ F_1 = P\tan\theta = P\theta = \tfrac{1}{2}F_k = \tfrac{1}{4}kL\theta$$

$$\therefore\ P = \tfrac{1}{4}kL$$

(**6.**) A compressive load P is applied aligned with the axis of a column of length 1.4 m with pinned ends. The column has a constant rectangular cross section of dimensions $d \times 3d$. Calculate the smallest possible value for d if $P = 108$ kN. The material properties are: $E = 200$ GPa, $\sigma_y = 247$ MPa. Consider a factor of safety (FS) of 1.92. $FS = \frac{\text{failure stress}}{\text{actual stress}}$.

Assume $R_{trans} \leq \frac{L}{r}$, fully buckling

$$\sigma_{actual} = \frac{P_{crit}}{A} = \frac{P_{crit}}{3d^2}$$

$$\because r = \sqrt{\frac{I}{A}} = \sqrt{\frac{\frac{3d^4}{12}}{3d^2}} = \sqrt{\frac{d^2}{12}} = \frac{\sqrt{3}}{6}d$$

$$\therefore \sigma_{crit} = \frac{P_{crit}}{A} = \frac{\pi^2 EI}{L^2}\frac{1}{A} = \pi^2 E\left(\frac{r}{L}\right)^2 = \frac{\pi^2 E}{\left(\frac{L}{r}\right)^2} = \frac{\pi^2 E d^2}{12L^2}$$

$$\because \frac{\sigma_{crit}}{\sigma_{actual}} = 1.92 \quad \therefore \frac{\pi^2 E d^2}{12L^2} = \frac{1.92 P_{crit}}{3d^2} \quad \therefore d = 0.03013\ (m)$$

$$\therefore \frac{L}{r} = 161$$

$$\because R_{trans} = \left(\frac{2\pi^2 E}{\sigma_Y}\right)^{\frac{1}{2}} = 126.42$$

$\therefore \frac{L}{r} > R_{trans}$, assumption is correct

Answers

1. Shown in video on Panopto (link on BB).

$$P_{crit} = \frac{\pi^2 EI}{L^2}$$

$$\frac{P}{P_{crit}} = (\frac{2}{\pi}\cos^{-}1(\frac{e}{v_c}))^2$$

2.

$$v_{AB} = B\sin(\mu z)$$

$$v_{BC} = C(\cos(\frac{\mu z}{2}) + \tan(\frac{\mu z}{4})\sin(\frac{\mu z}{2}))$$

$$P_{crit} = \frac{24.2EI}{L^2}$$

3.

$$\frac{\mathrm{d}^2 v}{z^2} + \mu^2 v = -\frac{wLz}{2EI}(z^2 - zL - \frac{2}{\mu^2})$$

$$P_{crit} = \frac{\pi^2 EI}{L^2}$$

4.

$$P_{crit} = \frac{\pi^2 EI}{(0.5L)^2}$$

5.

$$P_{crit} = \frac{1}{4}kL$$

6.

$$d = 0.0301\text{m}$$